Linear Classifiers and Logistic Regression

4

Pre

CS 3244
Machine Learning

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Recap and Forecast

Naïve Bayes:

- Naïve: assumes independence between input features
- Bayes: estimates posterior from prior, marginals and likelihoods
- Smoothing caters for unseen data

k NN:

- Instance-based non-parametric
- Votes from nearest neighbor
- Larger k to smooth decision boundaries
- Problems with high dimensional data

Other important concepts for ML:

- Smoothing, Cost (Loss) Functions, Noisy Targets and the Curse of Dimensionality

Learning Outcomes for this week

- Describe the basic idea of linear classification;
- Investigate linear classification using the Perceptron learning algorithm (PLA);
- Understand how both linear and logistic regression works;
- Understand the mathematical proof on how PLA converges;

- Other important concepts:
  Gradient Descent and Non-linear mappings.

The Perceptron

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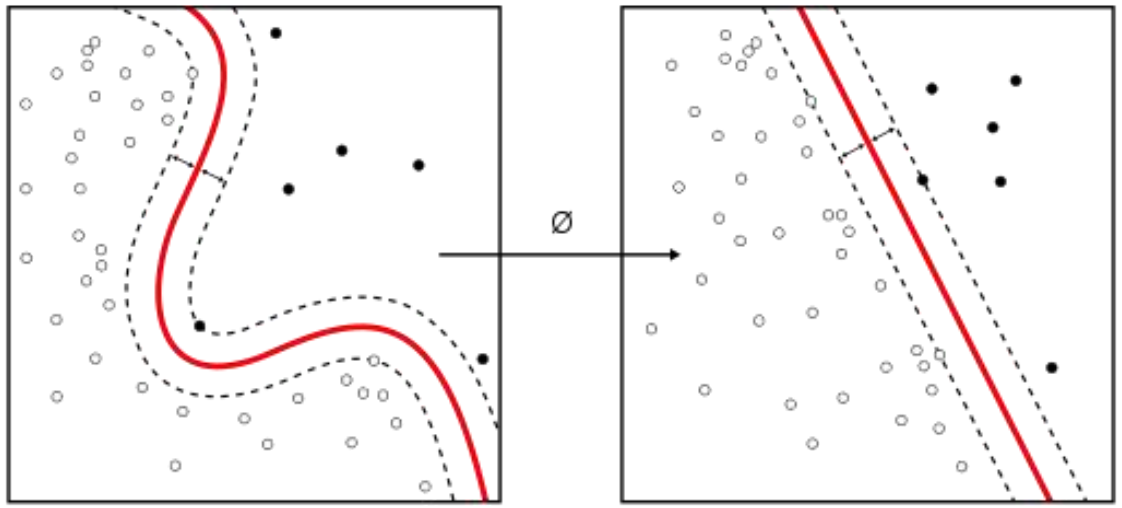
$\mathcal{H}$ for linear models

$\mathcal{H}$ represents the universe of possible hypotheses that a certain learning algorithm A can generate.

That is, given the choices of θ, what types of decision boundaries can it generate?



A ✗ B ✓

A simple hypothesis set – the perceptron

For input $\mathbf{x} = (x_1, x_2, \dots, x_n)$ *'attributes of a customer'*

- Approve credit if: $\sum_{i=1}^{n} \theta_i x_i > threshold$
- Deny credit otherwise: $\sum_{i=1}^{n} \theta_i x_i < threshold$

This linear formula $h \in \mathcal{H}$ can be written as:

$$h_\theta(x) = \text{sign}\left(\left(\sum_{i=1}^{n} \theta_i x_i\right) - threshold\right)$$

$$h(x) = \text{sign}((\sum_{i=1}^{n} \theta_i x_i) + \theta_0)$$

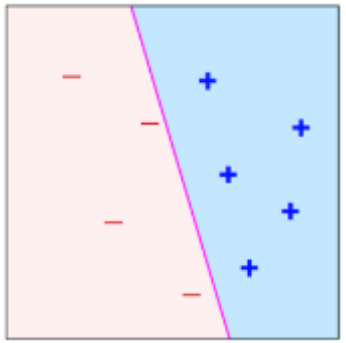
The 'bias weight' θ_0 corresponds to the threshold

Introduce an artificial coordinate $x_0 = 1$:

$$h_\theta(x) = \text{sign}(\sum_{i=0}^{n} \theta_i x_i)$$

In vector form, the perceptron implements

$$h(x) = \text{sign}(\theta^\top x)$$



Linearly separable data

A simple learning algorithm: PLA

The perceptron implements $h_\theta(\mathbf{x}) = \text{sign}(\theta^\top \mathbf{x})$

First initialize the weight vector θ in any fashion

Given the training set

$$\left((\mathbf{x}^{(1)}, y^{(1)}), (\mathbf{x}^{(2)}, y^{(2)}) \ldots, (\mathbf{x}^{(m)}, y^{(m)})\right)$$

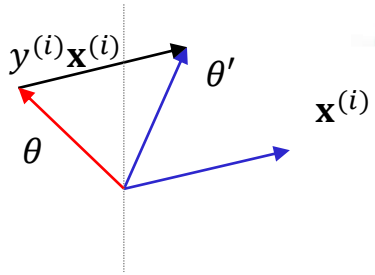
Pick a misclassified point $(\mathbf{x}^{(i)}, y^{(i)})$

$$\text{sign}(\theta^\top \mathbf{x}^{(i)}) \neq y^{(i)}$$

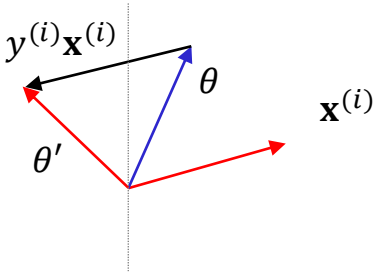
And update the weight vector

$$\theta' \leftarrow \theta + y^{(i)}\mathbf{x}^{(i)}$$

$y^{(i)} = +1$



$y^{(i)} = -1$



Iterations of PLA

One iteration of PLA:

$$\theta' \leftarrow \theta + y\mathbf{x}$$

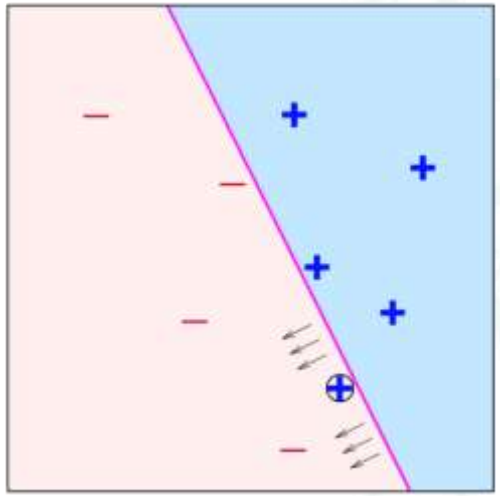
where $(\mathbf{x}, y)$ is a misclassified training point.

$t \leftarrow t + 1$

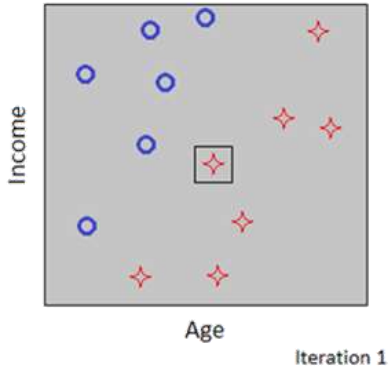
At iteration $t = 1, 2, 3, \ldots$, pick a misclassified point from:

$$\left(\left(\mathbf{x}^{(1)}, y^{(1)}\right), \left(\mathbf{x}^{(2)}, y^{(2)}\right) \ldots, \left(\mathbf{x}^{(m)}, y^{(m)}\right)\right)$$

and run a PLA iteration on it.



Does PLA work?



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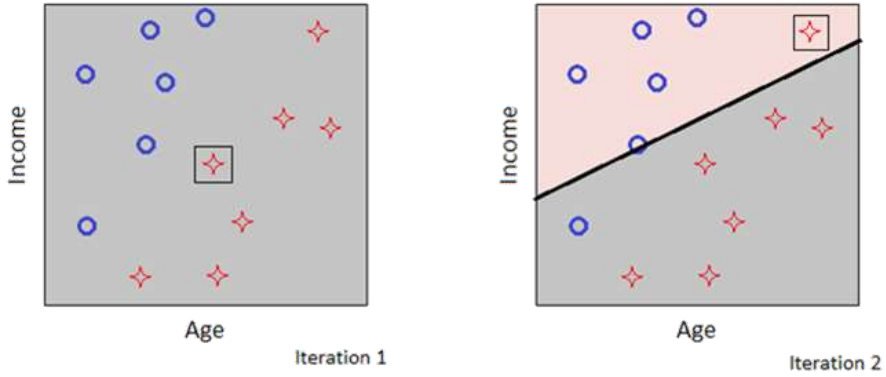

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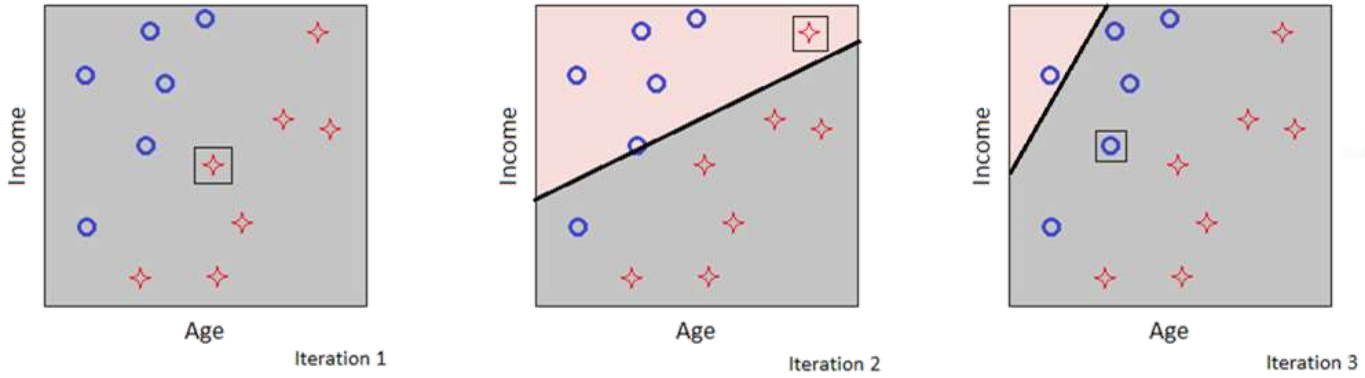

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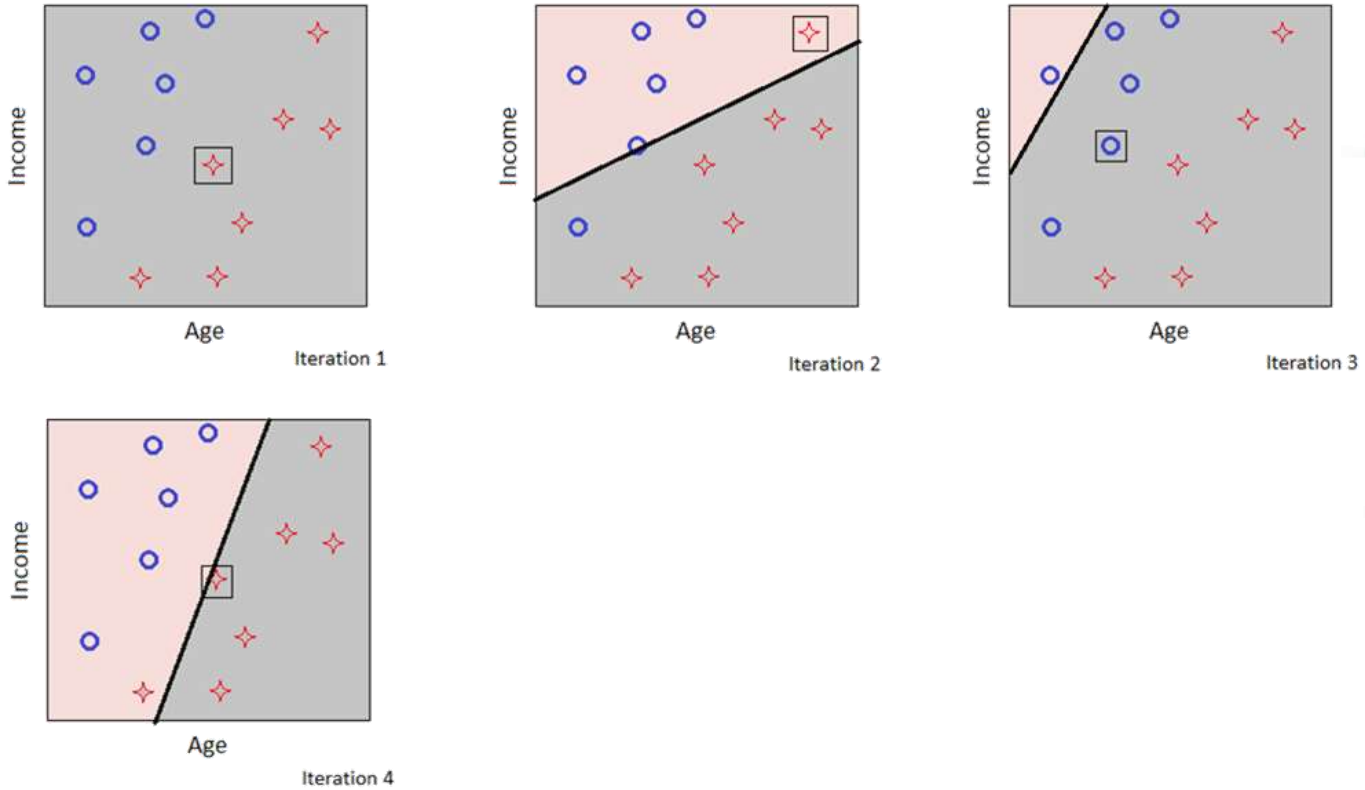

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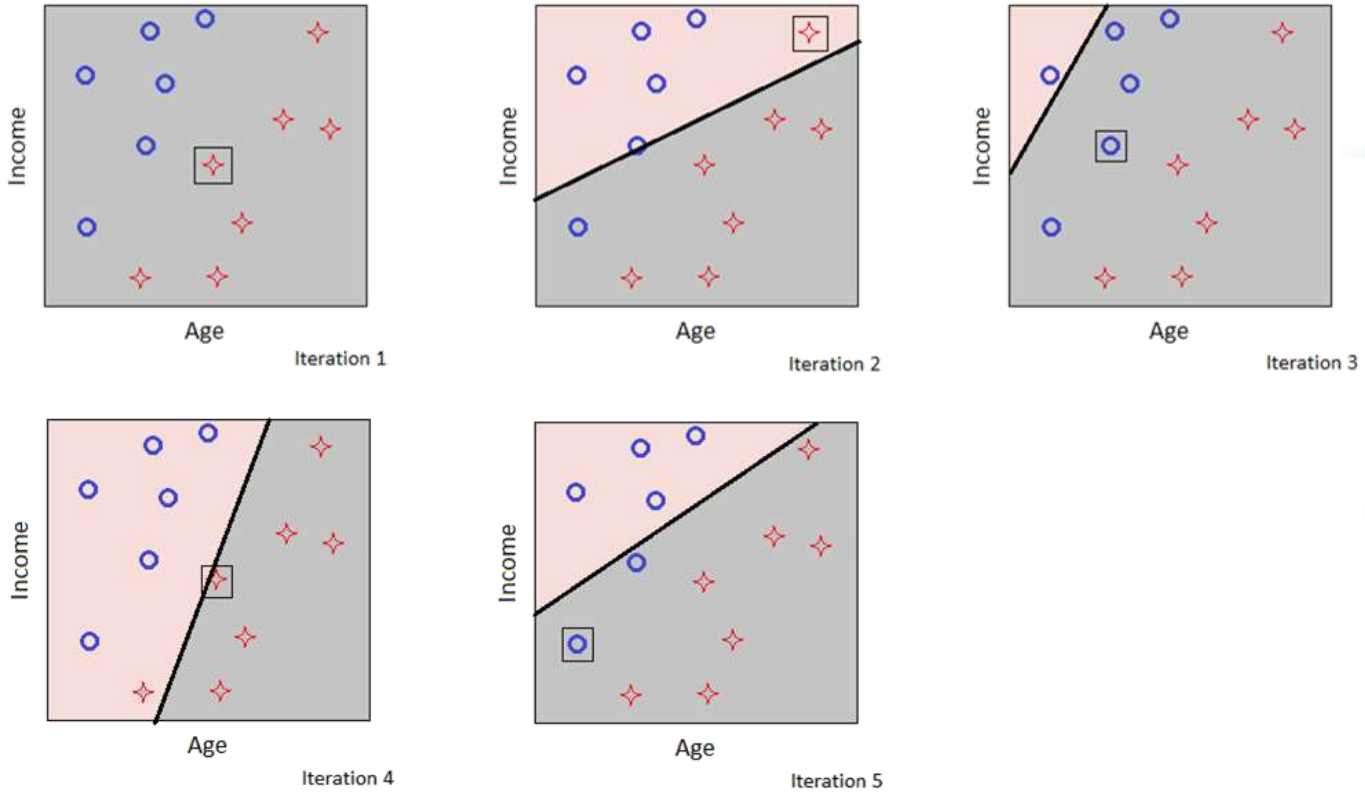

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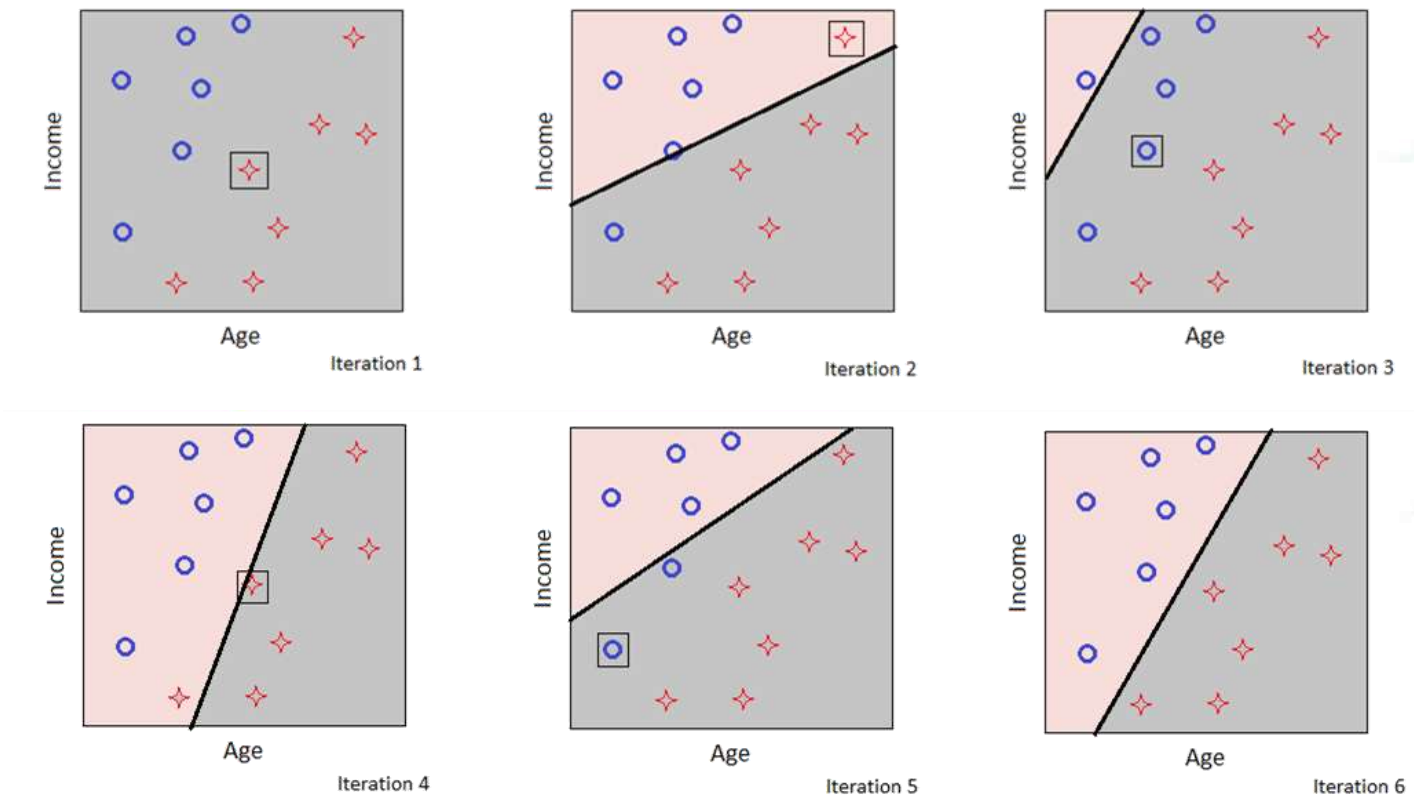
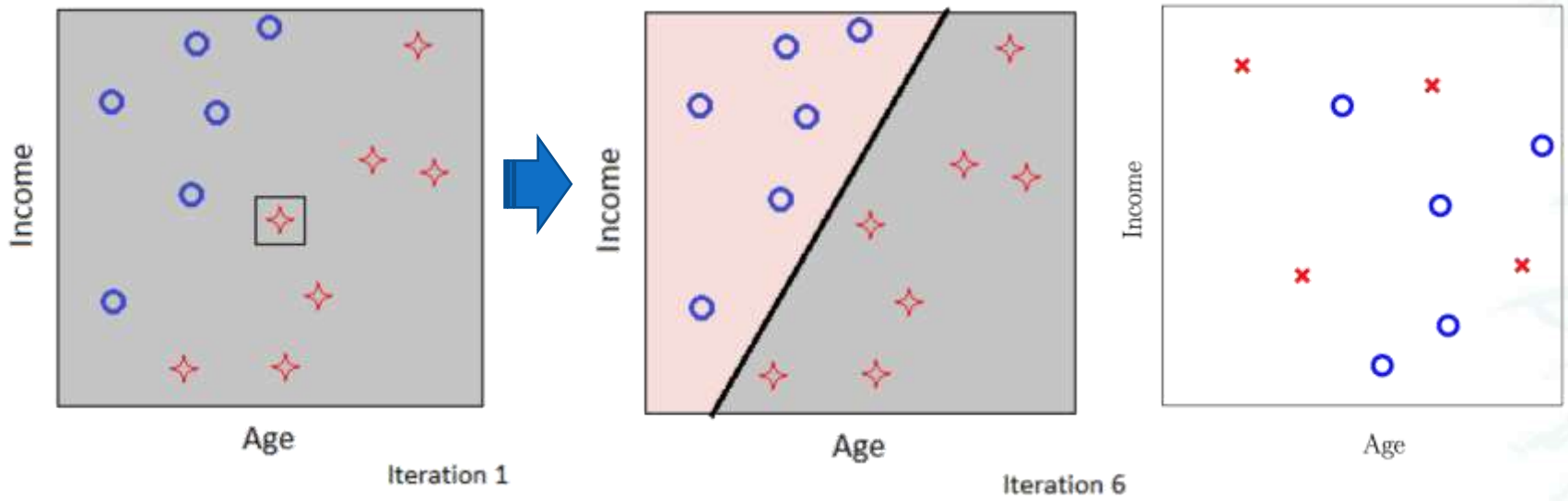


Does PLA work?

<u>Theorem</u>. If the data can be fit by a linear separator, then after some finite number of steps, PLA will find one.



Your Turn: but what if the perceptron cannot fit the data?

Summary – PLA

Perceptron Learning Algorithm

One of many iterative learning algorithms

Binary classification algorithm

Uses the signal $\theta^\top \mathbf{x}$

Starts with a random or zero weight vector θ

Nudges θ to correct individual errors on misclassified instances

Converges given sufficient iterations (proof later)

Linear Models

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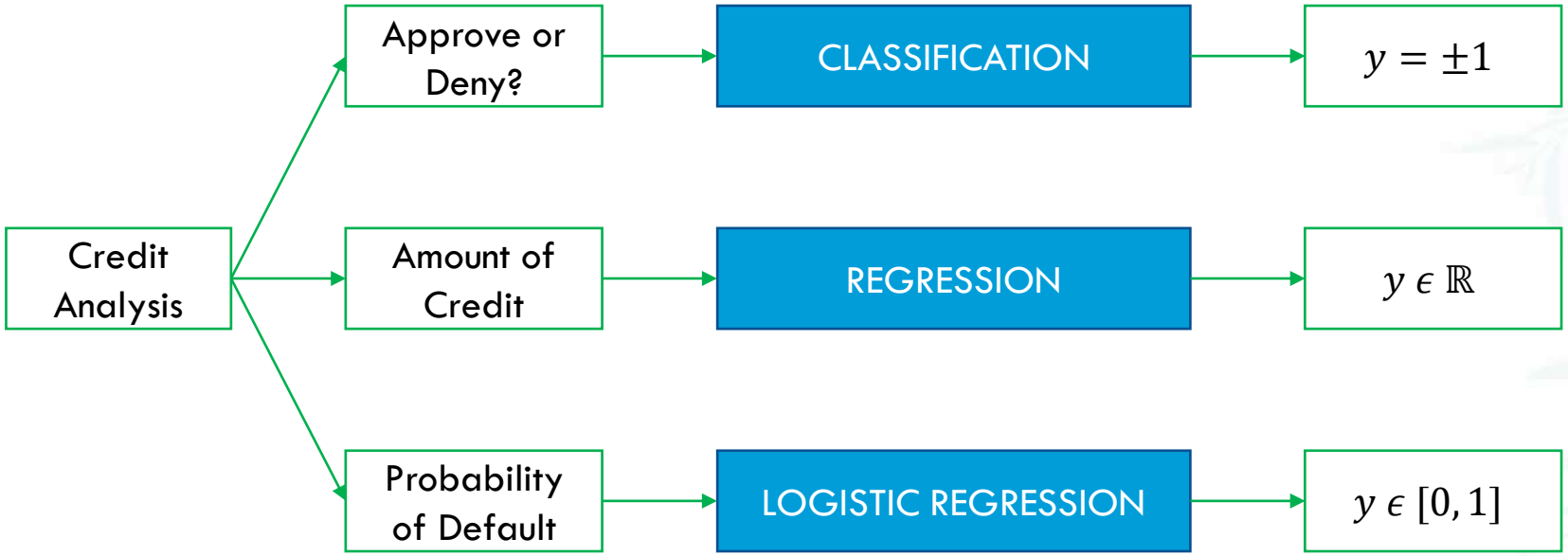
Three learning problems



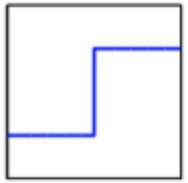
Linear models are the fundamental models.

The linear model is the first model to try.

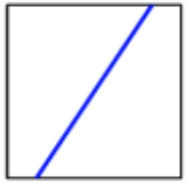
The linear signal

$$s = \theta^\top \mathbf{x}$$

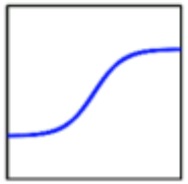
signal



$\text{sign}\,(\theta^\top \mathbf{x})$ $\quad y = \pm 1$

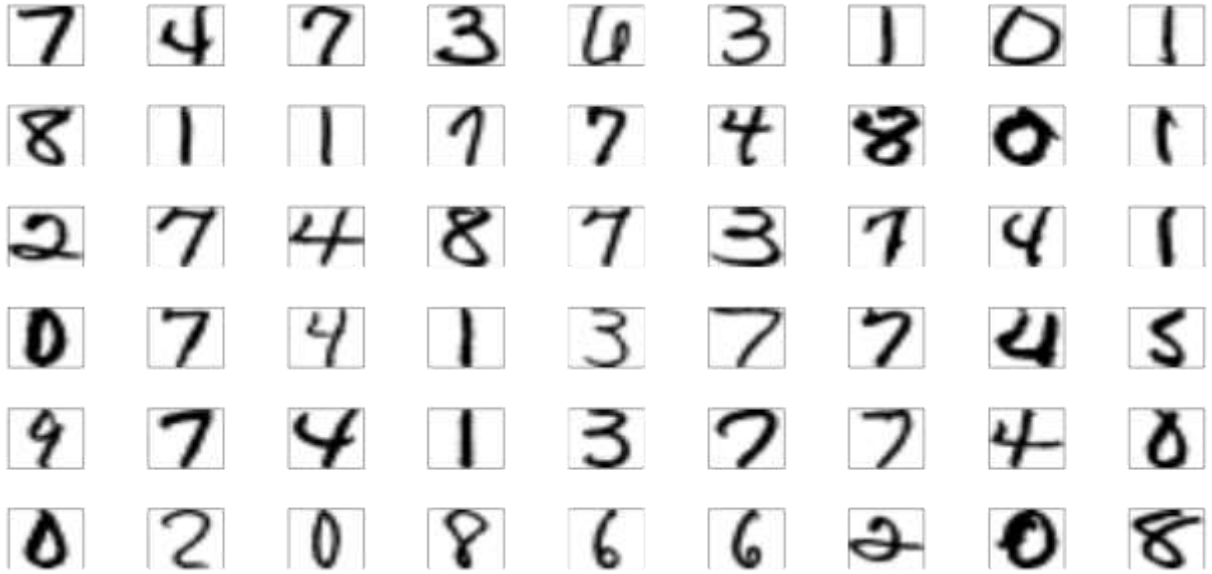


$\theta^\top \mathbf{x}$ $\quad y \in \mathbb{R}$ $\quad$ regression



$g(\theta^\top \mathbf{x})$ $\quad y \in [0, 1]$ $\quad$ probability

A real dataset



Each digit is a 16x16 pixel gray-intensity image.

[-1 -1 -1 -1 -1 -1 -1 -0.63 0.86 -0.17 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -0.99 0.3 1 0.31 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -0.41 1 0.99 -0.57 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -0.68 0.83 1 0.56 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -0.94 0.54 1 0.78 -0.72 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 0.1 1 0.92 -0.44 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -1 -0.26 0.95 1 -0.16 -1 -1 -1 -0.99 -0.71 -0.83 -1 -1 -1 -1 -1 -0.8 0.91 1 0.3 -0.96 -1 -1 -0.55 0.49 1 0.88 0.09 -1 -1 -1 -1 0.28 1 0.88 -0.8 -1 -0.9 0.14 0.97 1 1 1 0.99 -0.74 -1 -1 -0.95 0.84 1 0.32 -1 -1 0.35 1 0.65 -0.10 -0.18 1 0.98 -0.72 -1 -1 -0.63 1 1 0.07 -0.92 0.11 0.96 0.30 -0.88 -1 -0.07 1 0.64 -0.99 -1 -1 -0.67 1 1 0.75 0.34 1 0.70 -0.94 -1 -1 0.54 1 0.02 -1 -1 -1 -0.90 0.79 1 1 1 1 0.53 0.18 0.81 0.83 0.97 0.86 -0.63 -1 -1 -1 -1 -0.45 0.82 1 1 1 1 1 1 1 1 0.13 -1 -1 -1 -1 -1 -1 -0.48 0.81 1 1 1 1 1 1 0.21 -0.94 -1 -1 -1 -1 -1 -1 -1 -0.97 -0.42 0.30 0.82 1 0.48 -0.47 -0.99 -1 -1 -1 -1]

Input representation

Raw input $\mathbf{x} = (x_0, x_1, x_2, x_3, x_4, \dots, x_{256})$

Too many (257) parameters!

Linear model: $(\theta_0, \theta_1, \theta_2, \dots, \theta_{256})$

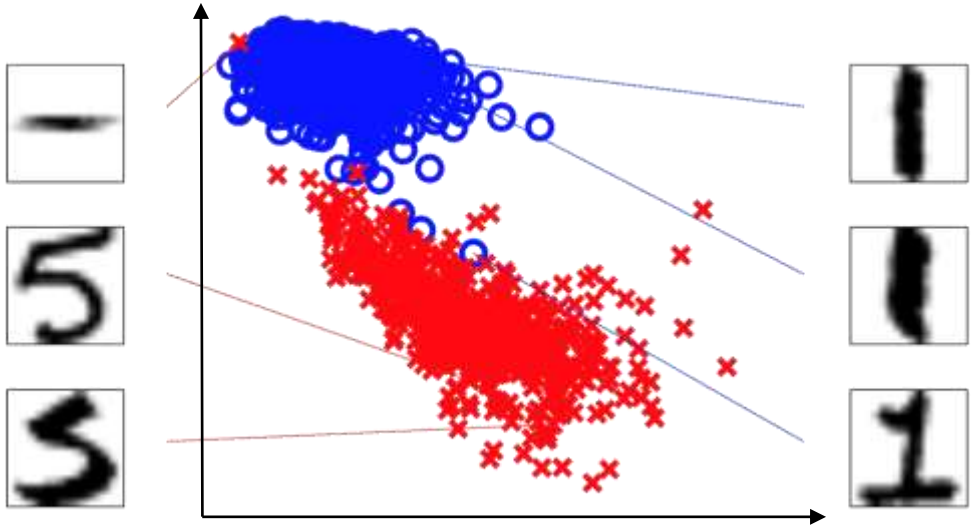
Features: extract useful information, e.g.,

Intensity and symmetry: $\mathbf{x} = (x_0, x_1, x_2)$

Linear model: $(\theta_0, \theta_1, \theta_2)$

Illustration of features

$\mathbf{x} = (x_0, x_1, x_2)$ $x_1 =$ intensity $x_2 =$ symmetry



Your Turn: which axes is which?

Pocket

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Iterations of PLA

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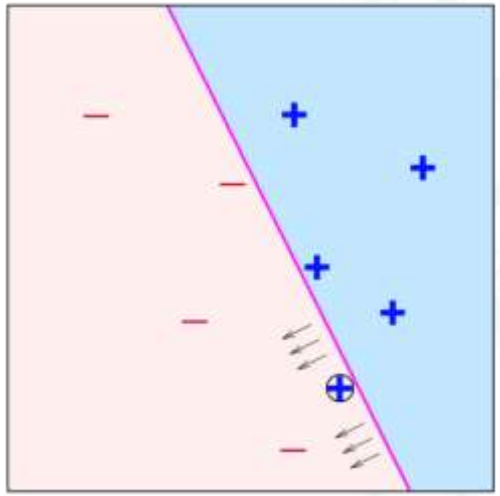
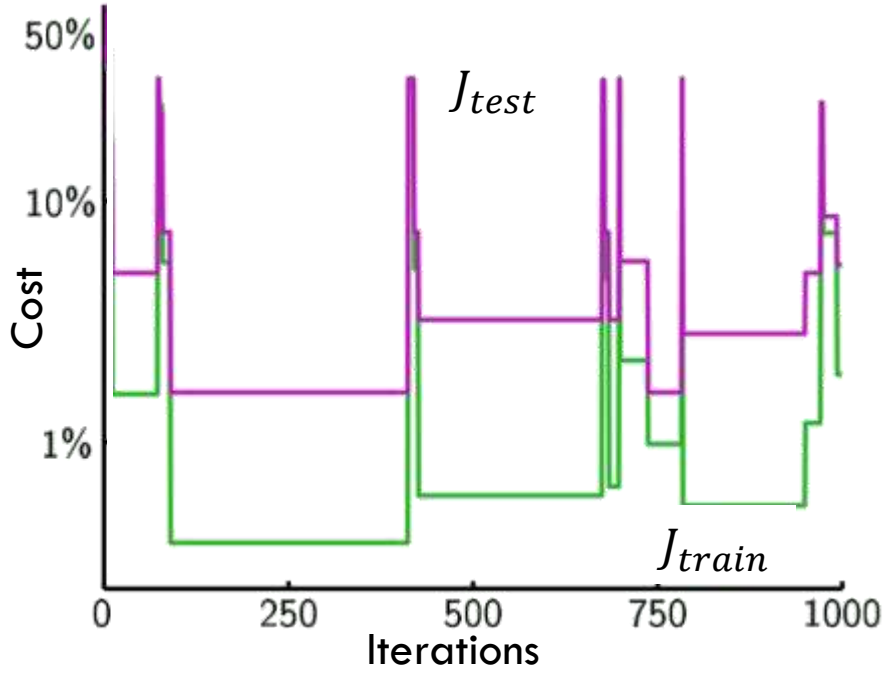
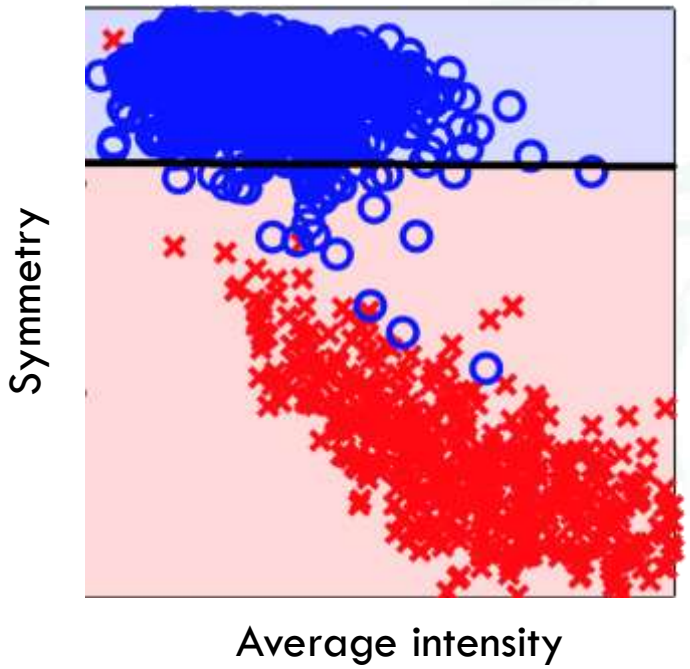


What PLA does

Evolution of J_{train} and J_{test}.

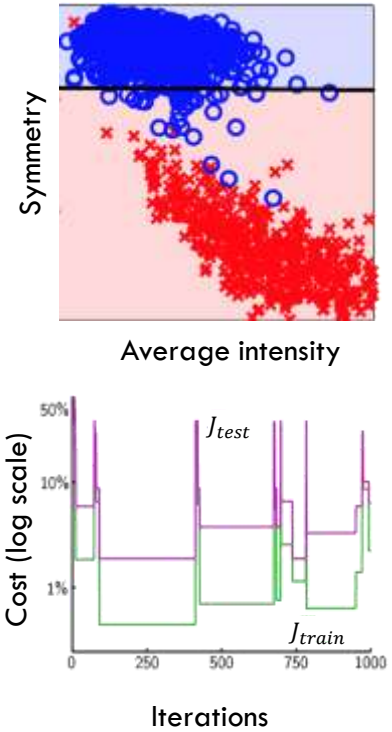


Final perceptron boundary



The pocket algorithm

PLA (for comparison):

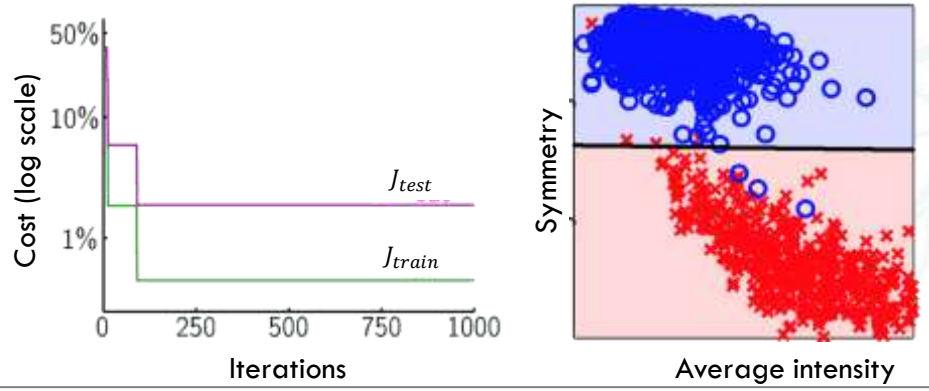


Pocket:

Run PLA

- At each step, keep the best J_{train} (and θ) so far

(it's not rocket science, but it works!)



Linear Regression

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Credit Approval

How much credit do we extend this person?

Criterion	Value
Age	32 years
Gender	Male
Salary	40 K
Debt	26 K
…	…
Years in Job	1 year
Years at Current Residence	3 years

~~Classification: Approve/Deny~~

- Regression: Credit line (real-valued dollar amount)
- Input: <*table on the left*>
- Linear regression output: $h_\theta(\mathbf{x}) = \sum_{i=0}^{n} \theta_i x_i = \theta^\top \mathbf{x}$
  $= \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_n x_n$
- Officers decide on credit lines based on historical data $\mathbf{X}$:
  $(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \ldots, (x^{(m)}, y^{(m)})$

$y^{(j)} \in \mathbb{R}$ is the credit line for customer $x^{(j)}$;
regression tries to replicate this.

Artwork credits: Project DebtRelief

$\mathbb{R}$-valued cost function

How well does $h_\theta(\mathbf{x}) = \theta^\top \mathbf{x}$ approximate $f(\mathbf{x})$?

In linear regression, we use *squared error*: $(h_\theta(\mathbf{x}) - f(\mathbf{x}))^2$

Training error:

$\mathbb{R}$-valued cost function

How well does $h_\theta(\mathbf{x}) = \theta^\top \mathbf{x}$ approximate $f(\mathbf{x})$?

In linear regression, we use *squared error*: $(h_\theta(\mathbf{x}) - f(\mathbf{x}))^2$

Training error:

$$J_{train}(h) = \frac{1}{m}\sum_{j=1}^{m}\left(h_\theta\left(x^{(j)}\right) - y^{(j)}\right)^2$$

Average

How bad is our prediction?

Illustration of linear regression

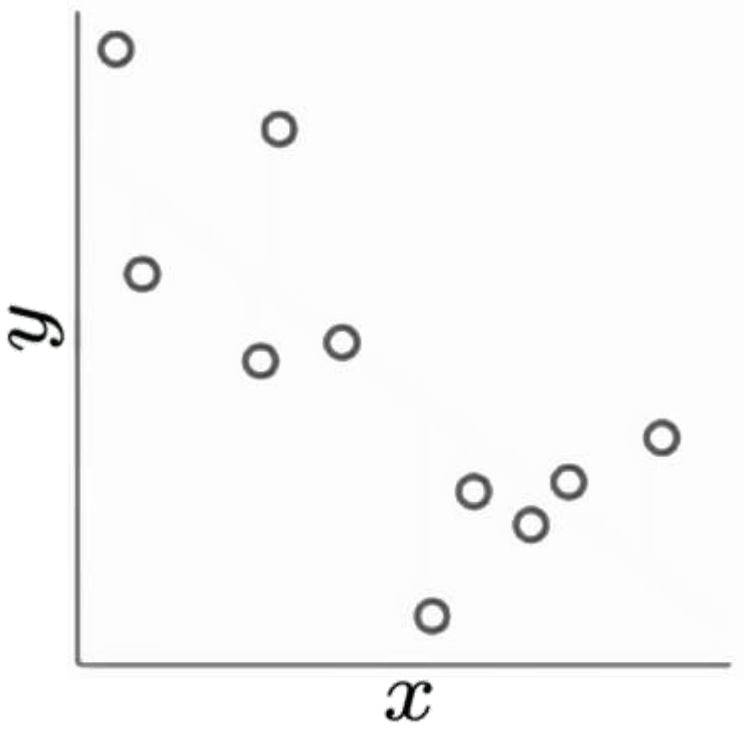
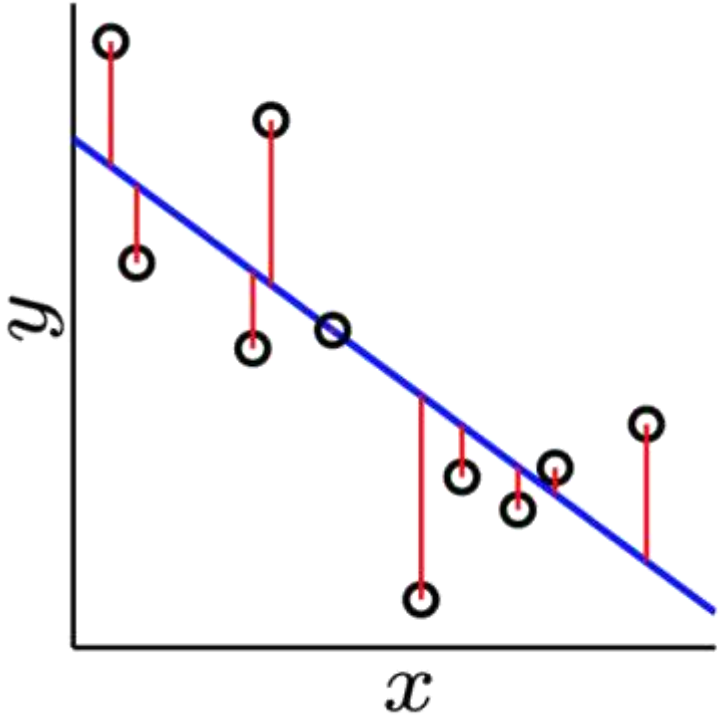
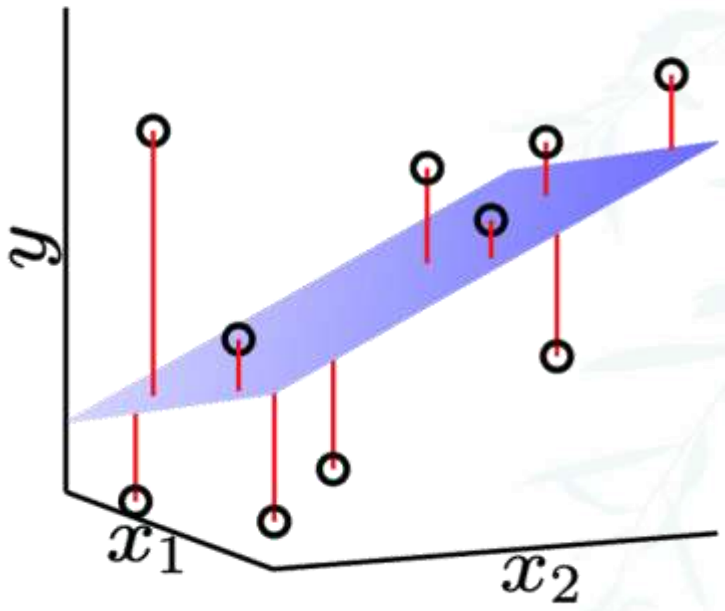


Illustration of linear regression





Hypothesis: $h_\theta(x) = \theta_0 + \theta_1 x$

Logistic Regression

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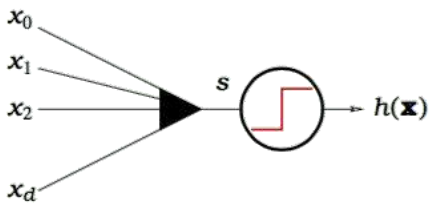
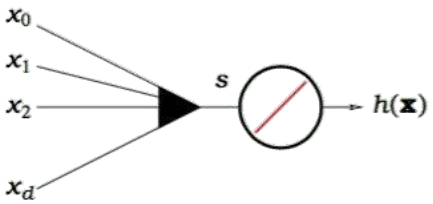
A third linear model

$$s = \sum_{i=0}^{n} \theta_i x_i$$

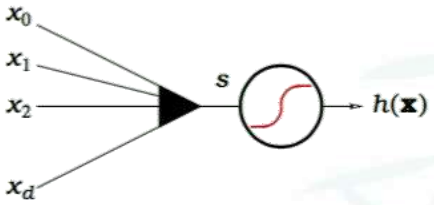
Linear classification



Linear regression



Logistic regression



The logistic function g

We choose a form for $g(\cdot)$ out of convenience.
We use the logistic function, which has the form:

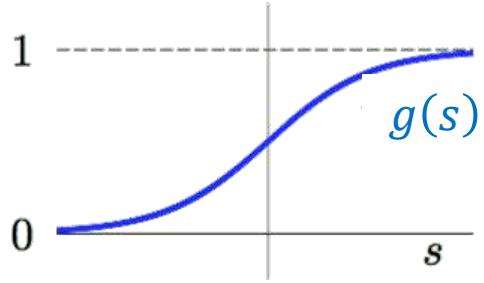
$$g(s) = \frac{exp^s}{1 + exp^s}$$

The logistic function g

We choose a form for $g(\cdot)$ out of convenience.
We use the logistic function, which has the form:

$$g(s) = \frac{exp^s}{1 + exp^s}$$

$g(s)$: needs to map any x: $\mathbb{R}$ to y: $[0,1]$



Soft threshold: uncertainty

Sigmoid: flattened out 's' shape

Probability Interpretation

$h_\theta(\mathbf{x}) = g(s)$ is interpreted as a probability

Logistic regression $\equiv y \in [0,1]$

Example: *Prediction of heart attacks*

Input $\mathbf{x}$: cholesterol level, age, weight, diabetic, etc.

output $g(s)$: Likelihood of a heart attack

The signal $s = \theta^\top \mathbf{x}$

"risk score"

Photo credits: Wikipedia - Chinese Buffet

Labels are binary, yet our outputs are probabilities

$$\mathbf{X} = \left(x^{(1)}, y^{(1)} = \pm 1\right), \ldots \left(x^{(m)}, y^{(m)} = \pm 1\right)$$

$x^{(j)}$ – a person's health information

$y^{(j)} = \pm 1$ – did they have a heart attack?

Why?

We cannot measure a <u>probability</u>.

We can only see the occurrence of an event and try to <u>infer</u> a probability.

Genuine Probability

Data $(\mathbf{x}, y)$ with binary y, generated by a noisy target:

$$P(y|x) = \begin{cases} f(\mathbf{x}) & \text{for } y = +1; \\ 1 - f(\mathbf{x}) & \text{for } y = -1. \end{cases}$$

The target $f: \mathbb{R}^n \to [0,1]$ is the probability

Learn $h_\theta(\mathbf{x}) = g(\theta^\top \mathbf{x}) \approx f(\mathbf{x})$

Cost Function for Logistic Regression

For logistic regression,

$$J_{train}(\theta) = \frac{1}{m}\sum_{j=1}^{m} ln(1 + exp^{-y^{(j)}\theta^{\top}x^{(j)}})$$

Iterative Solution

Compare to linear regression:

$$J_{train}(\theta) = \frac{1}{m}\sum_{j=1}^{m}\left(\theta^{\top}x^{(j)} - y^{(j)}\right)^2$$

Closed-form Solution; Iteration also possible

Gradient Descent

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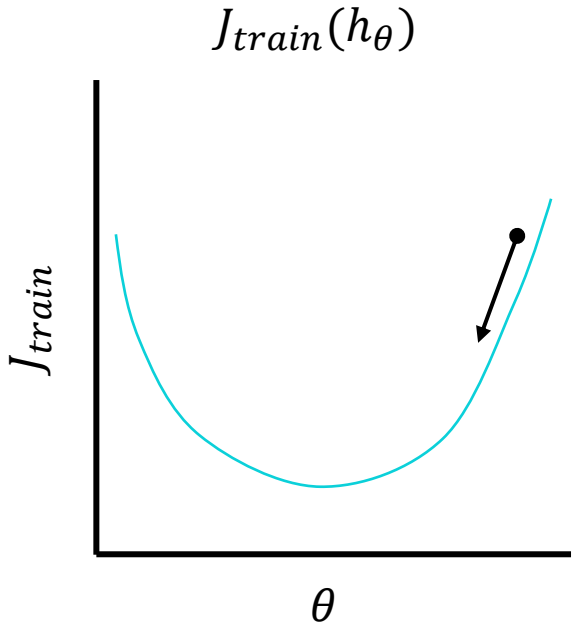
Climbing up (down) one step at a time

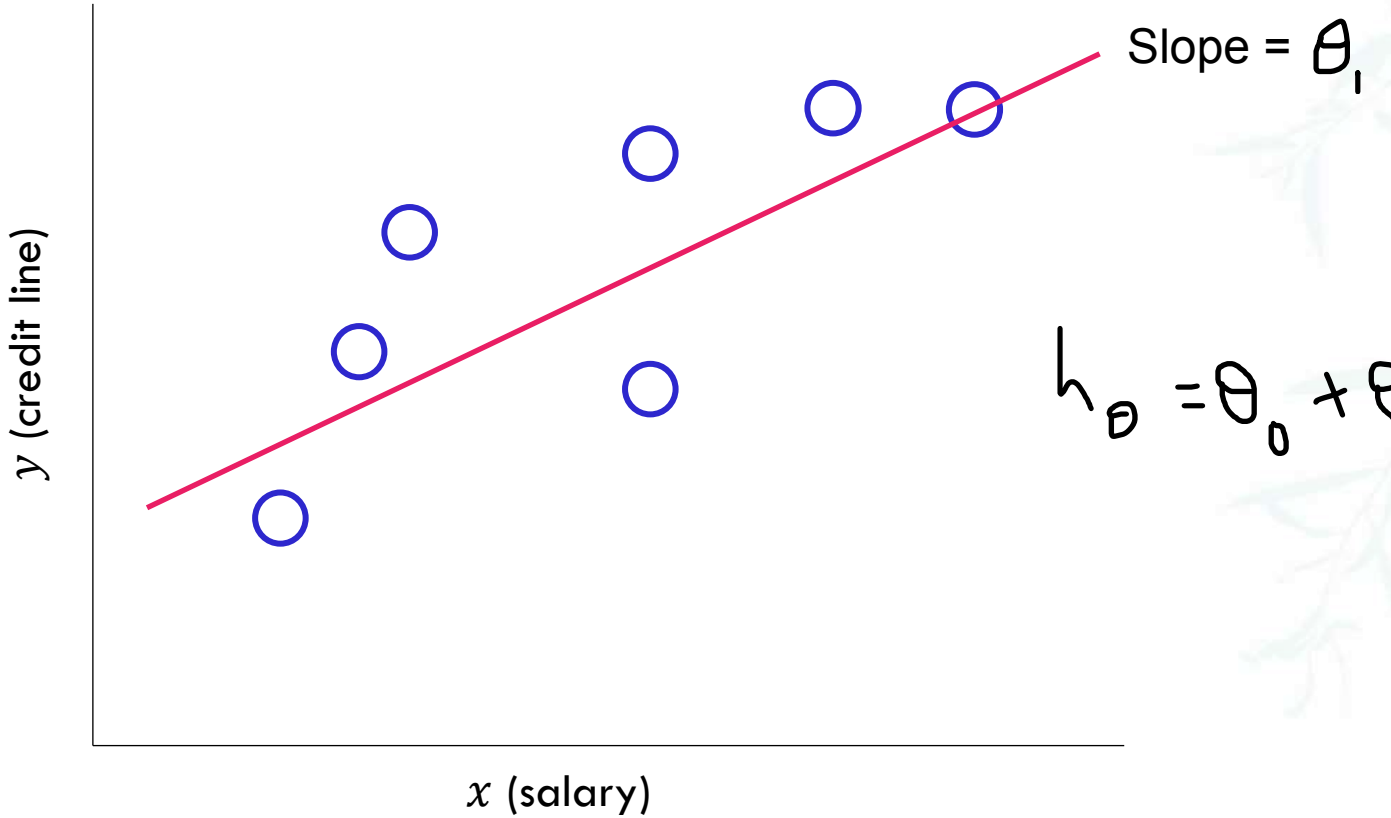
Photo credits: Wikimedia

Our J_{train} only has one valley



… Because J_{train} is a convex function of θ.

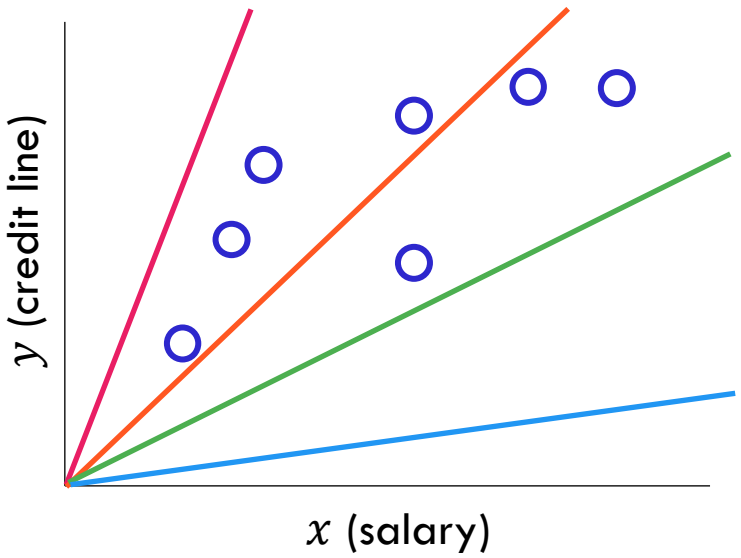
Univariate Linear Regression: Salary to predict Credit Line

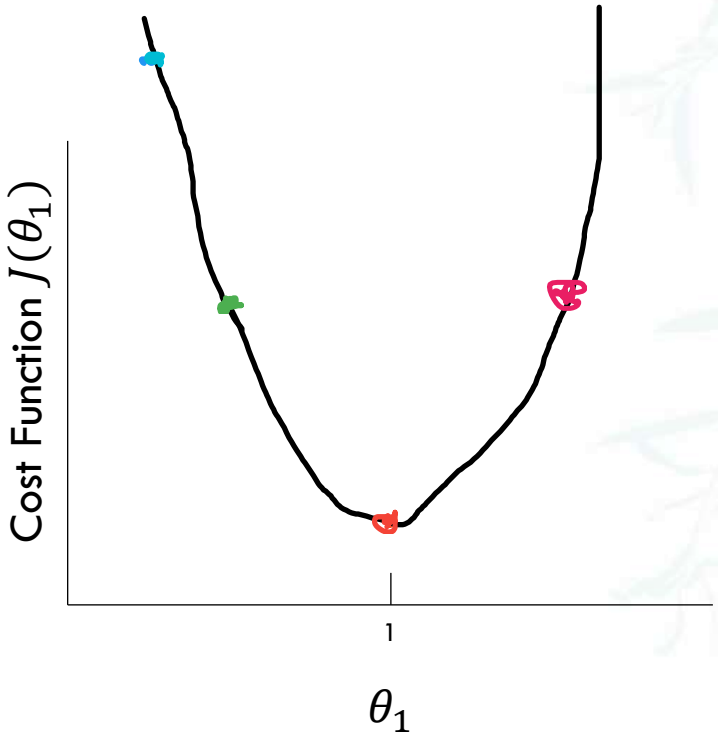


$$h_\theta = \theta_0 + \theta_1 x$$

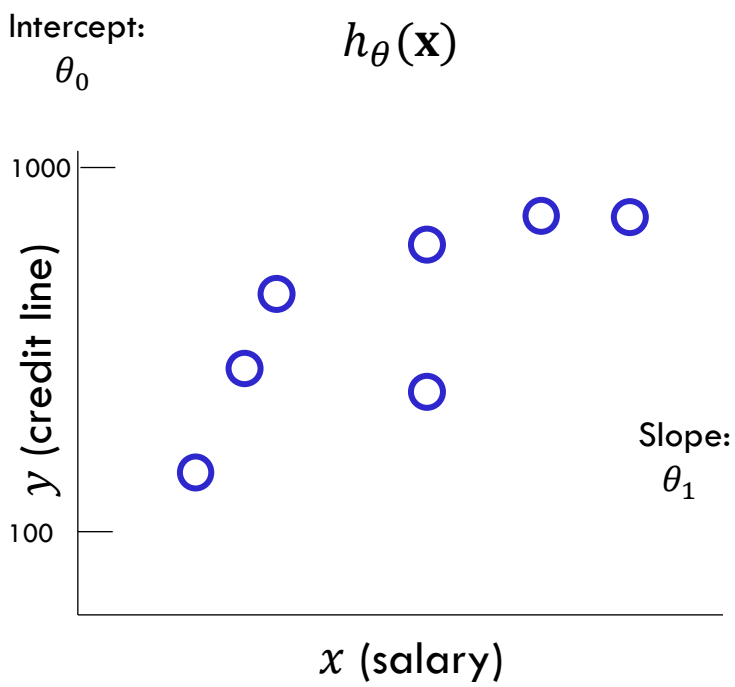
Ignoring the bias term

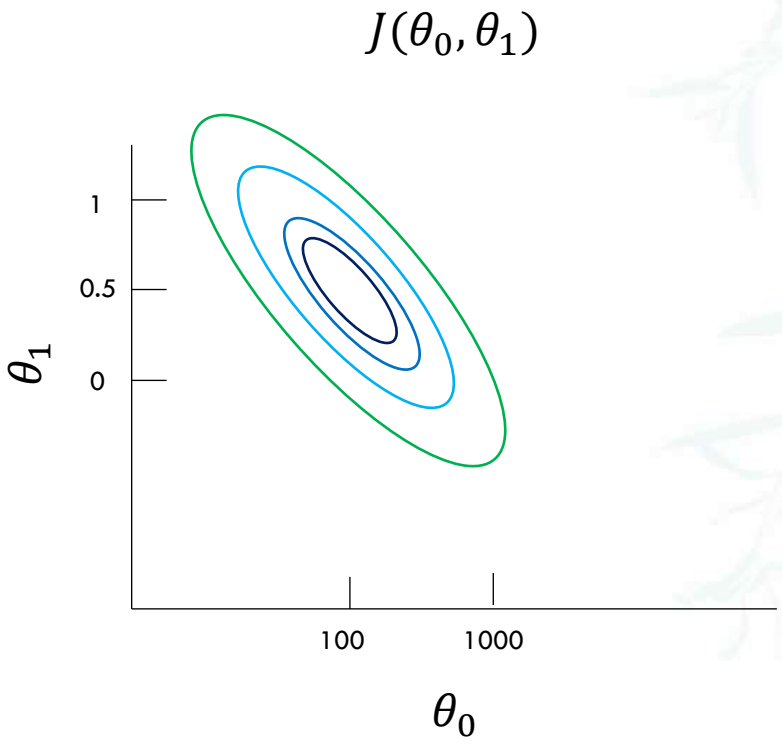
$h_\theta = \cancel{\theta_0} + \theta_1 x$





Cost Function $J(\theta_0, \theta_1)$





Iterative method: gradient descent

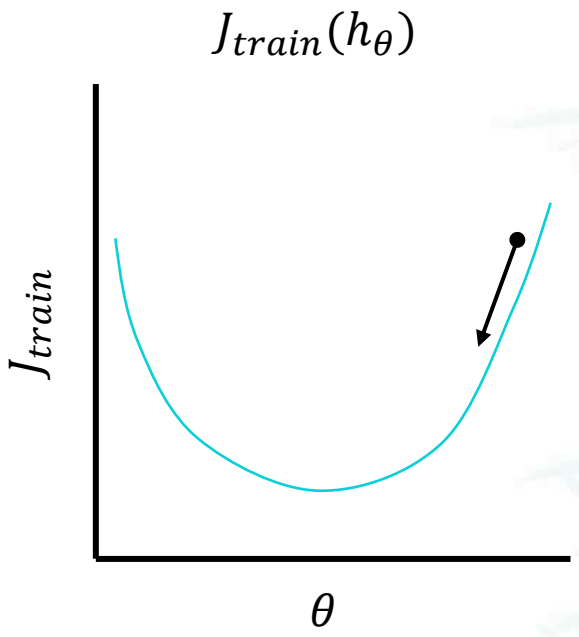
General method for nonlinear optimization

Start at $\theta(0)$; take a step along steepest slope

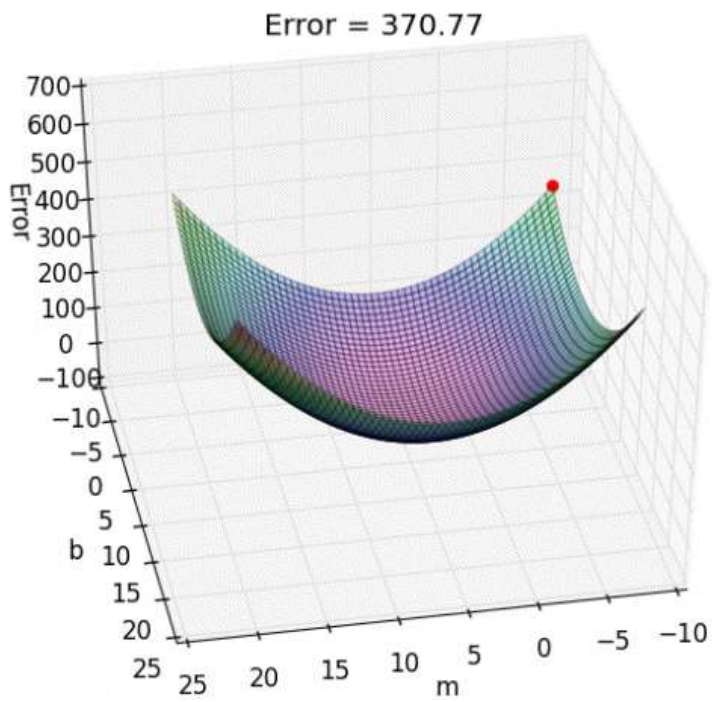
Fixed step size η : $\theta(1) = \theta(0) + \eta v$

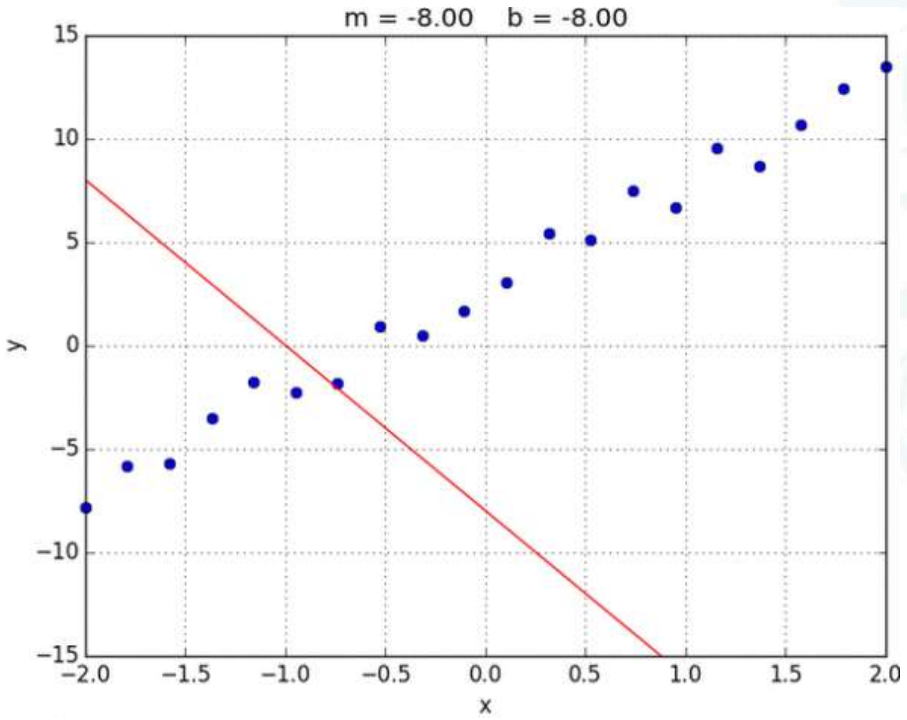
We get to pick the direction of v.
What is the best direction to pick?

The one to minimize J_{train}.



Gradient descent can minimize any smooth function.
Oh yeah!





Credits: Alykhan Tejani's Medium Post

Formula for the direction v

$$\Delta J_{train} = J_{train}\big(\theta(t+1)\big) - J_{train}\big(\theta(t)\big)$$

$$= J_{train}(\theta(t) + \eta v) - J_{train}\big(\theta(t)\big)$$

$$= \eta \nabla J_{train}(\theta(t)^{\top}) v \quad + O(\eta^2)$$

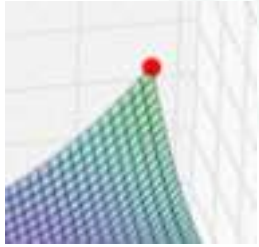
$$\geq -\eta \left\| \nabla J_{train}\big(\theta(t)\big) \right\|$$

Taylor approximation

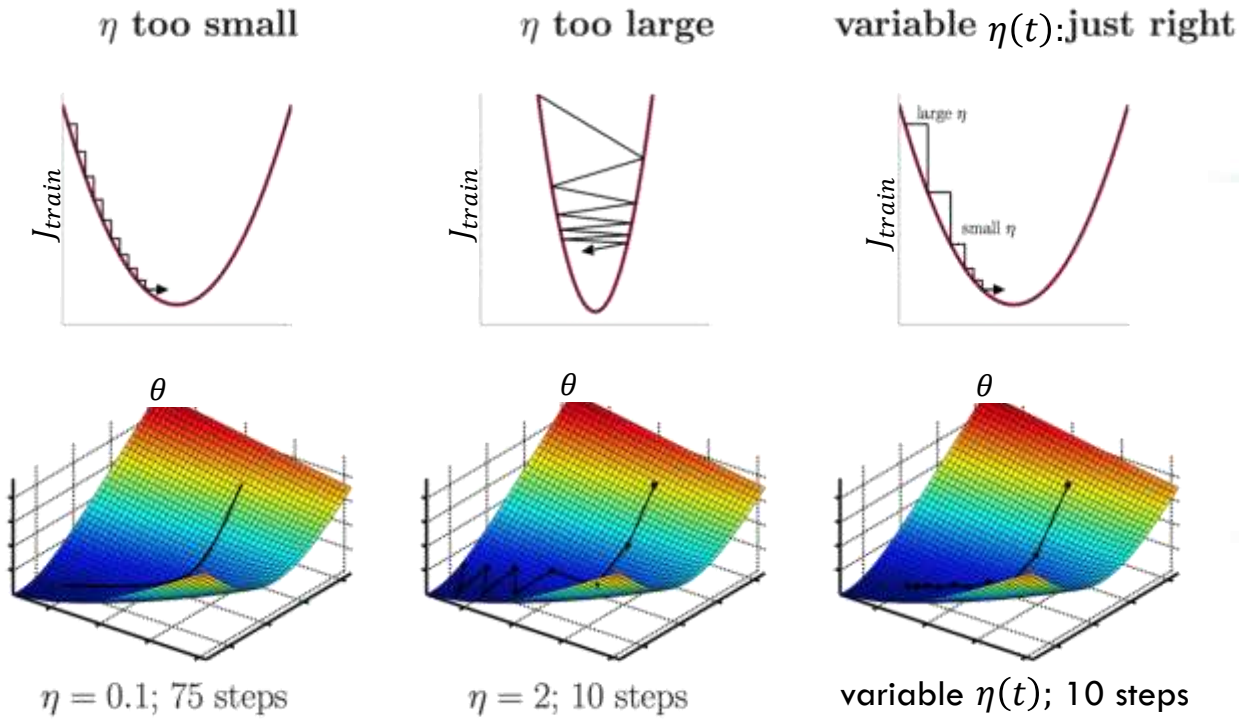
$f(x_1) = f(x_0) + \nabla f(x_0) \cdot (x_1 - x_0) + O(\|x_1 - x_0\|^2)$

Create v to be a unit vector:

$$v = -\frac{\nabla J_{train}(\theta(\mathrm{t}))}{\|\nabla J_{train}(\theta(t))\|}$$



Goldilocks step size



η should increase with the slope

Varying step size, for free

Instead of

$$\Delta\theta = \eta v$$

$$= -\eta \frac{J_{train}(\theta(t))}{\|J_{train}(\theta(t))\|}$$

Have

$$\Delta\theta = -\alpha J_{train}(\theta(t))$$

Fixed learning <u>rate</u> α instead of fixed learning <u>step</u> η

Logistic regression algorithm

1. Initialize the weights at $t = 0$ to $\theta(0)$
2. Do
3. Compute the gradient

$$\nabla(t) = \nabla J_{train}(\theta(t)) = -\frac{1}{m}\sum_{j=1}^{m}\frac{y^{(j)}x^{(j)}}{1 + e^{y^{(j)}\theta(t)^\top x^{(j)}}}$$

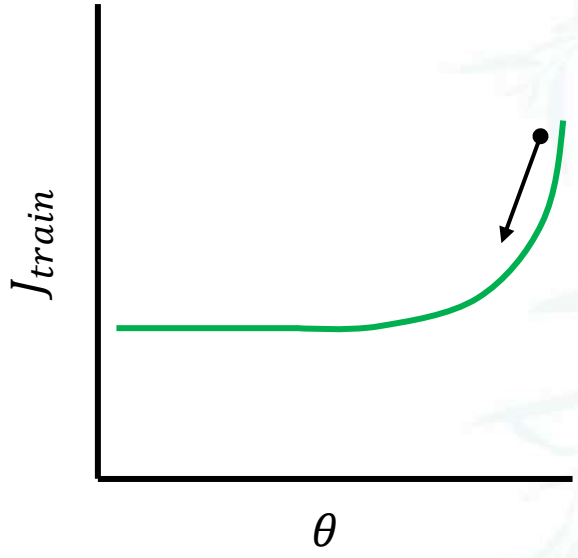
4. // Move in the direction $v(t) = -\nabla(t)$
   Update the weights $\theta(t+1) = \theta(t) - \alpha\nabla J_{train}$
5. Continue to next iteration, until it is <u>time to stop</u>
6. Return the final weights θ^*

Termination Condition

When to stop?

Natural choice: *gradient* < *threshold*

But lots of flat regions in most spaces:



Instead, use criteria:

1. error change is small and/or;
2. error is small;
3. maximum number of iterations is reached.